

David Zhao

Toronto, ON | 647-561-6811 | zhaol599@mylaurier.ca | LinkedIn | GitHub | Portfolio

EDUCATION

Wilfrid Laurier University

Bachelor of Science in Data Science(co-op) - 4.0 Program GPA

Waterloo, ON

Sept 2024 - Present

EXPERIENCE

Software Developer

Emergconnect

January 2026 – April 2026

Waterloo, ON, Canada

- **Optimized an NLP classification pipeline**, reducing patient complaint **misclassification rate by 25% across 5 triage categories**, by refactoring model logic and implementing systematic **prompt refinement and error analysis**
- **Built automated evaluation pipelines** using **test case design and confusion matrix analysis**, identifying edge cases across 200+ patient samples and refining classification thresholds to improve **prediction consistency by 30%**
- **Resolved mobile and web application bugs** through **root-cause analysis**, cutting **resolution time by 20%**

Instructional Assistant

Wilfrid Laurier University

September 2025 – Present

Waterloo, ON, Canada

- **Improved student project outcomes by 20%** by teaching **data pipeline workflows in Python, R, and SQL**, and guiding end-to-end data visualization using **ggplot2 and Matplotlib**
- **Communicated complex technical concepts** (**hypothesis testing, regression, ANOVA**) to **60+ students**, developing structured **Python and R** coding walkthroughs to improve **assignment completion rates**
- **Mentored 15+ students** on implementing **ML models** (linear regression, decision trees), bridging theory with hands-on coding practice

Software Developer

TeachTrack AI

May 2025 – August 2025

Toronto, ON, Canada

- **Engineered responsive frontend interfaces and authentication workflows**, reducing onboarding time by **30%**, by building reusable **React components and JavaScript** modules across core application pages
- **Optimized relational database architecture**, improving query performance by **40%**, by redesigning table schemas and indexing strategies to enhance **data accuracy and accessibility**
- **Integrated predictive analytics and statistical models** to improve personalized teaching support.

PROJECTS

A.U.R.A. - 2D floorplan to 3D environment - Winner of Bearhack | Vultr, Cloud Storage, Three.js, API April 2026

- **Engineered a three-agent AI pipeline** (Geometry Extraction, Placement Planning, Scene Assembly) using **Backboard APIs**, transforming **2D floor plans** into **structured data** and fully **interactive 3D environments**
- **Architected and deployed a full-stack application** on **Vultr** (VPS, Object Storage, PostgreSQL, Load Balancer), enabling **scalable scene generation, persistent storage, and real-time job tracking** for end-to-end workflows
- **Collaborated in a cross-functional team** using **GitHub** to design **intuitive user experiences** and align **AI outputs** with real-world spatial needs, ensuring the product was **robust and practically usable** for diverse use cases

CareSync - The Care Coordination OS - Winner HackCanada | Node.js, Auth0, React, Javascript March 2026

- **Built an AI-powered care coordination platform** by implementing **persistent patient memory** and automated briefing generation, streamlining communication between PSWs, coordinators, and families
- **Developed a multi-agent system** where automated agents detect **medication discrepancies** through nightly monitoring, generate caregiver handoff briefings, and draft personalized family updates from clinical notes
- **Implemented secure authentication and access control** to protect sensitive healthcare data

Spectra - AI monitoring App - Winner UOttHack | Next.js, Solana Web3.js, Supabase, PostgreSQL January 2026

- **Built a real-time security and analytics platform** for autonomous AI agents on **Solana**, by engineering **sub-second latency on-chain monitoring** to detect and prevent treasury drains and rogue execution, enabling safer deployments
- **Engineered an anomaly detection system** that flags suspicious behavior by analyzing **transaction frequency and spending velocity** across live agent activity
- Built a **WebGL dashboard** in **Next.js 14** for real-time agent monitoring and rogue behavior detection

ShopSmart - Full Stack E-commerce Web app | Python, HTML, React, Typescript, API August 2025

- **Developed a responsive e-commerce web app** with product listings, shopping cart, and secure checkout
- **Built a scalable backend API** to handle user authentication, order processing, and database management

TECHNICAL SKILLS AND CERTIFICATIONS

Certificates: Harvard CS50

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript/TypeScript, HTML/CSS, R

Frameworks: React, Node.js, Vue, Flask, JUnit, Material-UI, FastAPI, three.js

Libraries: Chart.js, NumPy, Matplotlib, ggplot2, scikit-learn, TailwindCSS, Axios